



Functional Dependency Equivalence and Minimal Cover

Database Systems • Beginner-Friendly Interactive Lecture

Today's promise

We will start from zero: attributes, tables, keys, functional dependencies, equivalence, closure, and minimal cover.

Teaching mode

Short examples, student questions, step-by-step checking, and practice at the end.

For BS Computer Science students with zero prior knowledge

Learning Objectives

What students should be able to do by the end

- Explain what a functional dependency means in a table.
- Differentiate FD, FD set, and equivalent FD sets.
- Use Armstrong's axioms and common inference rules.
- Find attribute closure step by step.
- Check whether two FD sets are equivalent.
- Find a minimal cover by simplifying dependencies.
- Solve practice questions without memorizing blindly.

Simple target

Given rules like $A \rightarrow B$ and $B \rightarrow C$, students should be able to say what else must be true.

Exam target

Given two FD sets, students should be able to prove equivalent or not equivalent using closure.

Normalization target

Minimal cover is used before decomposition into normal forms such as 3NF.

Class Activity / Think: Ask students: What makes a database table “well organized”? Let them answer before giving theory.

Why Dependencies Matter

Dependencies describe the real meaning inside a table

Real problem

If the same CourseID appears many times, its CourseName should not randomly change in different rows.

Database rule

CourseID → CourseName means: if CourseID is fixed, CourseName must also be fixed.

Why we care

Dependencies help us remove redundancy, avoid update errors, find keys, and normalize tables.

StudentID	StudentName	CourseID	CourseName
101	Ali	DB101	Database Systems
102	Sara	DB101	Database Systems
103	Hamza	AI201	Artificial Intelligence

Observation: DB101 always gives Database Systems.

So we can write: CourseID → CourseName

Class Activity / Think: From the table, can StudentID determine StudentName? Why?

Basic Concepts: Attribute and Relation



Before dependencies, students must know the table vocabulary

Attribute

A column of a table. Example: StudentID, Name, Department.

Tuple / Row

One record in a table. Example: one student record.

Relation / Table

A collection of rows and columns. Example: STUDENT(StudentID, Name, DeptID).

StudentID	Name	DeptID	DeptName
S01	Ali	D1	CS
S02	Sara	D2	SE
S03	Ayan	D1	CS

Relation schema: STUDENT(StudentID, Name, DeptID, DeptName)

Class Activity / Think: Point to one attribute, one row, and the full relation in the table.

Basic Concepts: Keys and Attribute Types



Keys help us identify rows and understand dependencies

Super Key

Any attribute set that can uniquely identify a row. It may contain extra attributes.
Example: {StudentID, Name}

Candidate Key

A minimal super key. No extra attribute can be removed.
Example: {StudentID}

Primary Key

The selected candidate key used by the database designer.
Example: StudentID

Prime Attribute

An attribute that is part of some candidate key.
Example: StudentID.

Non-prime Attribute

An attribute that is not part of any candidate key.
Example: Name, DeptName.

- Key idea: keys identify rows.
- Functional dependency idea: one attribute value determines another attribute value.
- These ideas connect during normalization.

Class Activity / Think: Can {StudentID, Name} be a candidate key if StudentID alone is unique? Why not?

Quick Introduction to Functional Dependency



The basic rule: one side determines the other side

$X \rightarrow Y$

Read as: “X determines Y”

Left side = determinant.
Right side = dependent
attribute(s).

Meaning

If two rows have the same X value, then they must have the same Y value.

Student example

StudentID $\rightarrow$ StudentName
Same StudentID cannot have two different names in a clean table.

Course example

CourseID $\rightarrow$ CourseTitle
Same CourseID should always represent the same course title.

Class Activity / Think: In Employee(EmpID, EmpName, DeptID, DeptName), which FD looks natural?

FD, FD Set, and Equivalent FD Sets

These three terms are different

Functional Dependency

One rule.

Example: $A \rightarrow B$

FD Set

A collection of rules.

Example: $F = \{ A \rightarrow B, B \rightarrow C \}$

Equivalent FD Sets

Two sets that can derive exactly the same dependencies.

Example: F and G have same meaning.

Example: $F = \{ A \rightarrow B, B \rightarrow C \}$
 $G = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$

Why G may still be equivalent

$A \rightarrow C$ is already implied by $A \rightarrow B$ and $B \rightarrow C$. So G only writes an extra rule that F can already produce.

Class Activity / Think: Does adding a dependency always change the meaning of an FD set? Discuss.

Armstrong's Axiom 1: Reflexivity

A rule that is always true because the right side is already included in the left side

If $Y \subseteq X$, then $X \rightarrow Y$

Simple meaning

If you know a bigger set of attributes, you also know any smaller part of it.

Example

$\{\text{StudentID}, \text{CourseID}\} \rightarrow \text{StudentID}$
Because StudentID is already inside the left side.

Not exciting, but useful

Reflexivity helps prove many closure results, especially when attributes are already present.

Class Activity / Think: Is $\{A, B, C\} \rightarrow B$ always true? Why?



Armstrong's Axiom 2: Augmentation

If a rule is true, adding the same extra attribute to both sides keeps it true

If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Simple meaning

If X can determine Y, then X with extra information can still determine Y with that same extra information.

Example

If $\text{StudentID} \rightarrow \text{StudentName}$, then
 $\text{StudentID, CourseID} \rightarrow \text{StudentName, CourseID}$.

Think of it like this:

If ID tells the name, then ID + course tells the name + course.

Class Activity / Think: From $A \rightarrow B$, can we infer $AC \rightarrow BC$?

Armstrong's Axiom 3: Transitivity

Dependency can pass through a middle attribute

If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

Simple meaning

If X determines Y, and Y determines Z, then X indirectly determines Z.

Course example

CourseID $\rightarrow$ Instructor
Instructor $\rightarrow$ OfficeRoom
So CourseID $\rightarrow$ OfficeRoom.



Therefore: CourseID $\rightarrow$ OfficeRoom

Class Activity / Think: From RollNo $\rightarrow$ DeptID and DeptID $\rightarrow$ DeptName, what can RollNo determine?

Additional Inference Rules

Rules that make derivation easier

Union

If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$.

Example: $A \rightarrow B$ and $A \rightarrow C$ means $A \rightarrow BC$.

Decomposition

If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$.

Example: $A \rightarrow BC$ means $A \rightarrow B$ and $A \rightarrow C$.

Pseudo-transitivity

If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$.

Example: $A \rightarrow B$ and $CB \rightarrow D$ means $CA \rightarrow D$.

Why this matters

When checking equivalence, we need to see whether one FD set can generate the dependencies of another FD set.

- Union helps combine right sides.
- Decomposition helps split right sides.
- Pseudo-transitivity helps with combined left sides.

Class Activity / Think: From $A \rightarrow B$ and $A \rightarrow C$, which rule gives $A \rightarrow BC$?

Attribute Closure: The Main Tool

Closure tells us everything an attribute set can determine

Definition

The closure of X , written X^+ , is the set of all attributes that can be determined from X using the given FDs.

Simple analogy

Start with a keychain. Each dependency is a locked door. If you have the needed key, you can open the door and collect more keys.

Algorithm for X^+

1. Start with $X^+ = X$.
2. Check each FD in the set.
3. If the left side is inside X^+ , add the right side to X^+ .
4. Repeat until no new attribute can be added.

Use closure for

Finding keys

Checking whether $X \rightarrow Y$ is implied

Checking whether two FD sets are equivalent

Class Activity / Think: Why do we repeat the process instead of checking each FD only once?

Attribute Closure: Step-by-Step Example

Example: Find A^+ under $F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

$$F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$$

Step	Current closure	Reason
Start	$A^+ = \{A\}$	We start with A.
1	$A^+ = \{A, B\}$	$A \rightarrow B$, so add B.
2	$A^+ = \{A, B, C\}$	$B \rightarrow C$, and B is now available.
3	$A^+ = \{A, B, C, D\}$	$C \rightarrow D$, and C is now available.
Stop	$A^+ = \{A, B, C, D\}$	No new attribute can be added.

Result

A determines A, B, C, and D. So $A \rightarrow D$ is implied even if it was not directly written.

Class Activity / Think: Can we say A is a candidate key for R(A,B,C,D)? What condition must be true?

Using Closure to Find Keys

A key must determine all attributes in the relation

Given relation

$R(A, B, C, D)$
 $F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

Find A^+

$A^+ = \{A, B, C, D\}$
So A determines all attributes.

Conclusion

A is a super key. Because A has only one attribute, it is also a candidate key.

Check another attribute

$B^+ = \{B, C, D\}$. It does not include A, so B is not a super key.

Rule of thumb: If X^+ contains all attributes of R, then X is a super key. If no attribute can be removed from X, then X is a candidate key.

Class Activity / Think: For $R(A,B,C,D)$, is C a key under the same FD set? Calculate C^+ mentally.

Functional Dependency Equivalence

Two FD sets are equivalent if they have the same meaning

Definition

FD sets F and G are equivalent if every dependency implied by F is also implied by G , and every dependency implied by G is also implied by F .

Simple analogy

Two different recipe cards are equivalent if they produce the same final dish, even if the wording is different.

Equivalence means:

$$F^+ = G^+$$

Important warning

Do not check equivalence by only comparing the written dependencies. Use closure to check derivability.

Class Activity / Think: Can two FD sets have different number of dependencies and still be equivalent?

Yes. Two FD sets can have different numbers of dependencies and still be equivalent.

Because equivalence means:

Both FD sets can derive the same information.

It does **not** mean both sets must have the same number of FDs.

Example

Suppose:

$F = \{ A \rightarrow B, A \rightarrow C \}$

$G = \{ A \rightarrow BC \}$

How to Check FD Equivalence

A practical exam method using closure

1. Take every FD $X \rightarrow Y$ in F .
2. Compute X^+ using G .
3. If Y is inside X^+ , then G can derive that FD.
4. Repeat for all FDs in F .
5. Now take every FD in G and compute closure using F .
6. If both directions pass, then F and G are equivalent.

One-direction check

If every FD in F can be derived from G , then F is covered by G .

Two-direction check

For equivalence, both F covered by G and G covered by F must be true.

Fast question

To test $X \rightarrow Y$ using F , compute X^+ under F . If $Y \subseteq X^+$, the dependency is implied.

Class Activity / Think: Why is checking only one direction not enough?

Solved Example: Equivalent FD Sets



Check if $F = \{ A \rightarrow B, B \rightarrow C \}$ and $G = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$ are equivalent

Relation: $R(A, B, C)$

Check F using G

$A \rightarrow B$ is in G.
 $B \rightarrow C$ is in G.
So every FD of F is covered by G.

Check G using F

$A \rightarrow B$ is in F.
 $B \rightarrow C$ is in F.
For $A \rightarrow C$: A^+ under $F = \{A, B, C\}$. So $A \rightarrow C$ is derived.

Final answer: F and G are equivalent.

Lesson

An extra FD does not always mean extra meaning. It may only be a dependency that was already implied.

Class Activity / Think: Which rule helps us derive $A \rightarrow C$ from $A \rightarrow B$ and $B \rightarrow C$?

Solved Example: Not Equivalent FD Sets

A missing indirect rule can break equivalence

Relation: R(Course, Instructor, Room)

F

$F = \{ \text{Course} \rightarrow \text{Instructor}, \text{Instructor} \rightarrow \text{Room} \}$

G

$G = \{ \text{Course} \rightarrow \text{Instructor}, \text{Course} \rightarrow \text{Room} \}$

Check Instructor $\rightarrow$ Room using G

Instructor⁺ under G = {Instructor}

Room is not added. So G cannot derive Instructor $\rightarrow$ Room.

Final answer: F and G are NOT equivalent.

Class Activity / Think: What exact dependency exists in F but cannot be derived from G?

Minimal Cover: Introduction

A minimal cover is a simpler FD set with the same meaning

Definition

A minimal cover of F is a simpler FD set that is equivalent to F and has no unnecessary attributes or unnecessary dependencies.

Real-life analogy

Like rewriting messy lecture notes into clean short notes without losing any important meaning.

Example messy set: $F = \{ A \rightarrow BC, B \rightarrow C, A \rightarrow C \}$

Possible cleaner set: $F_c = \{ A \rightarrow B, B \rightarrow C \}$

Main condition

The minimal cover must still be equivalent to the original FD set.

Class Activity / Think: Why is it dangerous to remove a dependency without checking equivalence?

Why Minimal Cover Is Needed

Minimal cover prepares dependencies for normalization

- It removes repeated or unnecessary information from the FD set.
- It makes dependency checking easier.
- It helps in 3NF decomposition algorithms.
- It reduces confusion when many dependencies look similar.
- It preserves the same meaning as the original FD set.

Before minimal cover

$F = \{ A \rightarrow BC, A \rightarrow C, B \rightarrow C \}$
Messy and partly repeated.

After minimal cover

$F_c = \{ A \rightarrow B, B \rightarrow C \}$
Shorter but same meaning.

Strict rule

Shorter is not enough. It must be equivalent.

Class Activity / Think: Can a smaller FD set be wrong? Give a simple reason.

Minimal Cover Step 1: Split RHS

Each dependency should have only one attribute on the right side

Rule

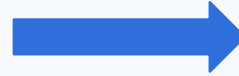
Replace $X \rightarrow YZ$ with $X \rightarrow Y$ and $X \rightarrow Z$.

Why

Single-attribute RHS makes later checking easier and standard.

Example:

$A \rightarrow BC$



$A \rightarrow B$
 $A \rightarrow C$

Another example

CourseID $\rightarrow$ Instructor, Room
becomes
CourseID $\rightarrow$ Instructor
CourseID $\rightarrow$ Room

Class Activity / Think: Split this FD: StudentID $\rightarrow$ Name, Department, Email

Minimal Cover Step 2: Remove Extraneous Attributes from LHS



An attribute is extraneous if the FD still works without it

Rule

In $AB \rightarrow C$, attribute B is extraneous if A alone can still determine C using the FD set.

Test

Replace $AB \rightarrow C$ with $A \rightarrow C$ temporarily, or compute A^+ using the adjusted set. If C appears, B can be removed.

Example: $F = \{ AB \rightarrow C, A \rightarrow B \}$

Question	Check	Result
Is B extraneous in $AB \rightarrow C$?	A^+ using $\{AB \rightarrow C, A \rightarrow B\}$: A gives B, then AB gives C.	Yes. $AB \rightarrow C$ can become $A \rightarrow C$.
Why?	A can first get B, so A has both A and B.	B was not needed on the left side.

Class Activity / Think: In $AB \rightarrow C$, what closure should we compute to test whether B is extraneous?

Minimal Cover Step 3: Remove Redundant Dependencies

A dependency is redundant if other dependencies already imply it

Rule

To test $X \rightarrow Y$, remove it temporarily. Then compute X^+ using the remaining FDs. If Y is still found, the FD is redundant.

Example

$F = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$
Is $A \rightarrow C$ redundant?

Check

Remove $A \rightarrow C$. Compute A^+ using $\{A \rightarrow B, B \rightarrow C\}$.
 $A^+ = \{A, B, C\}$. So $A \rightarrow C$ is still implied.

Conclusion

$A \rightarrow C$ is redundant and can be removed.
Minimal cover becomes $\{ A \rightarrow B, B \rightarrow C \}$.

Class Activity / Think: Which FD should be removed from $\{ \text{RollNo} \rightarrow \text{DeptID}, \text{DeptID} \rightarrow \text{DeptName}, \text{RollNo} \rightarrow \text{DeptName} \}$?

Full Minimal Cover Solved Example



Find minimal cover of $F = \{ A \rightarrow BC, B \rightarrow C, AB \rightarrow D, C \rightarrow D \}$

Step	Work	Result
1. Split RHS	$A \rightarrow BC$ becomes $A \rightarrow B$ and $A \rightarrow C$.	$\{A \rightarrow B, A \rightarrow C, B \rightarrow C, AB \rightarrow D, C \rightarrow D\}$
2. Remove extraneous LHS	In $AB \rightarrow D$, B is extraneous because A^+ gives B, C , and D .	$AB \rightarrow D$ becomes $A \rightarrow D$
3. Remove redundant FDs	$A \rightarrow C$ is redundant because $A \rightarrow B$ and $B \rightarrow C$. $A \rightarrow D$ is redundant because $A \rightarrow B \rightarrow C \rightarrow D$.	$\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$
Final	No RHS has multiple attributes. No unnecessary LHS attributes. No redundant FD remains.	Minimal cover = $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

Check the meaning

The final set is shorter, but it still gives $A \rightarrow B$, $A \rightarrow C$, $A \rightarrow D$, and $C \rightarrow D$. That is why it is valid.

Class Activity / Think: Why was $A \rightarrow D$ removed as redundant? Explain using a dependency chain.



Minimal Cover and Equivalence + Common Mistakes

Minimal cover must preserve the original meaning

Connection

A minimal cover F_c of F must be equivalent to F . This means $F^+ = F_c^+$.

Main test

After simplification, the new FD set must still derive the old dependencies.

- Mistake 1: Splitting the left side. Wrong. Split only the right side in Step 1.
- Mistake 2: Removing an attribute because it “looks extra” without closure test.
- Mistake 3: Checking only one direction for equivalence.
- Mistake 4: Forgetting to repeat closure until no new attribute appears.
- Mistake 5: Thinking minimal cover must be unique. Different valid minimal covers may exist.

Class Activity / Think: Which mistake do you think students usually make in exams: extraneous LHS or redundant FD?



Practice: Functional Dependency Equivalence

Solve these 5 questions before seeing the next slide

Instructions

For each question, decide whether F and G are equivalent. Use closure. Write at least one reason.

#	Relation	F	G	Task
1	R(A,B,C)	$\{A \rightarrow B, B \rightarrow C\}$	$\{A \rightarrow B, A \rightarrow C, B \rightarrow C\}$	Equivalent?
2	R(A,B,C,D)	$\{A \rightarrow BC, C \rightarrow D\}$	$\{A \rightarrow B, A \rightarrow C, C \rightarrow D, A \rightarrow D\}$	Equivalent?
3	R(Course,Instructor,Room)	$\{Course \rightarrow Instructor, Instructor \rightarrow Room\}$	$\{Course \rightarrow Instructor, Course \rightarrow Room\}$	Equivalent?
4	R(A,B,C)	$\{AB \rightarrow C, C \rightarrow B\}$	$\{A \rightarrow C, C \rightarrow B\}$	Equivalent?
5	R(A,B,C,D)	$\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$	$\{A \rightarrow B, B \rightarrow C, B \rightarrow D, C \rightarrow D\}$	Equivalent?

Student space

Use rough work: compute X⁺ under the opposite FD set when needed.

Class Activity / Think: Do not guess by counting FDs. Prove using closure.



Solutions: Functional Dependency Equivalence Practice

Answers for the 5 equivalence questions

#	Answer	Reason
1	Equivalent	$A \rightarrow C$ is implied by $A \rightarrow B$ and $B \rightarrow C$. Other FDs match.
2	Equivalent	$A \rightarrow BC$ splits into $A \rightarrow B$ and $A \rightarrow C$. $A \rightarrow D$ follows from $A \rightarrow C$ and $C \rightarrow D$.
3	Not equivalent	G cannot derive $\text{Instructor} \rightarrow \text{Room}$. Instructor^+ under $G = \{\text{Instructor}\}$.
4	Not equivalent	F cannot derive $A \rightarrow C$. A^+ under $F = \{A\}$.
5	Equivalent	$B \rightarrow D$ is implied by $B \rightarrow C$ and $C \rightarrow D$. The other FDs match.

Exam habit

When the answer is “not equivalent,” always mention one dependency that fails and show its closure.

Class Activity / Think: Which two questions were not equivalent, and what dependency failed in each?

Practice: Minimal Cover



Solve these 5 questions before seeing the next slide

Instructions

Find a minimal cover for each FD set. Show all steps: split RHS, remove extraneous LHS, remove redundant FD.

#	Relation	FD Set F	Task
1	R(A,B,C)	{A→BC, B→C}	Find minimal cover.
2	R(A,B,C,D)	{AB→C, A→C, C→D}	Find minimal cover.
3	R(A,B,C,D)	{A→B, B→C, A→C, C→D}	Find minimal cover.
4	R(A,B,C,D,E)	{AB→C, C→D, D→E, A→C}	Find minimal cover.
5	R(StudentID, CourseID, Instructor, Room, Grade)	{StudentID CourseID→Grade, CourseID→Instructor Room, Instructor→Room}	Find minimal cover.

Student space

Write your final answer as a set of single-RHS dependencies.

Class Activity / Think: Remember: split RHS, not LHS.

Solutions: Minimal Cover Practice



Answers for the 5 minimal cover questions

#	Minimal Cover	Key reason
1	$\{A \rightarrow B, B \rightarrow C\}$	After splitting $A \rightarrow BC$ into $A \rightarrow B$ and $A \rightarrow C$, $A \rightarrow C$ is redundant because $A \rightarrow B \rightarrow C$.
2	$\{A \rightarrow C, C \rightarrow D\}$	$AB \rightarrow C$ is redundant because $A \rightarrow C$ already gives C .
3	$\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$	$A \rightarrow C$ is redundant because $A \rightarrow B \rightarrow C$.
4	$\{A \rightarrow C, C \rightarrow D, D \rightarrow E\}$	$AB \rightarrow C$ is redundant because $A \rightarrow C$ already gives C .
5	$\{\text{StudentID CourseID} \rightarrow \text{Grade}, \text{CourseID} \rightarrow \text{Instructor}, \text{Instructor} \rightarrow \text{Room}\}$	Split $\text{CourseID} \rightarrow \text{Instructor Room}$; $\text{CourseID} \rightarrow \text{Room}$ is redundant through $\text{Instructor} \rightarrow \text{Room}$.

Final check

Every answer above has single-attribute RHS and preserves equivalence with the original set.

Class Activity / Think: Pick one solution and prove one removed dependency using closure.

Quick Revision

The whole lecture in one page

Functional Dependency

$X \rightarrow Y$ means same X value must have same Y value.

Closure

X^+ is everything X can determine using the FD set.

Equivalence

F and G are equivalent when both can derive each other's dependencies.

Minimal Cover Steps

1. Split RHS.
2. Remove extraneous LHS attributes.
3. Remove redundant FDs.

Best exam method

For derivation $X \rightarrow Y$, compute X^+ . If Y is inside X^+ , the FD is implied.

One strict warning

Never remove anything from an FD set by intuition only. Use closure to prove it.

Class Activity / Think: Exit ticket: In one sentence, explain why minimal cover must be equivalent to the original FD set.

Given Relation

$R(A, B, C, D, E)$

Given:

$F = \{ A \rightarrow B, B \rightarrow C, AC \rightarrow D, D \rightarrow E \}$

$G = \{ A \rightarrow B, B \rightarrow C, A \rightarrow D, D \rightarrow E \}$

Check whether F and G are equivalent.

Practice 2

$R(A, B, C, D)$

$F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

$G = \{ A \rightarrow B, B \rightarrow D, C \rightarrow D \}$

Practice 3

$R(A, B, C, D, E)$

$F = \{ A \rightarrow B, B \rightarrow C, A \rightarrow D, D \rightarrow E \}$

$G = \{ A \rightarrow B, B \rightarrow C, A \rightarrow DE \}$

Given Relation

$R(A, B, C, D, E)$

Given:

$F = \{ AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A \}$

$G = \{ AB \rightarrow D, D \rightarrow E, E \rightarrow A, AB \rightarrow C \}$

Check whether F and G are equivalent.

Practice 4

$R(A, B, C, D, E)$

$F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E \}$

$G = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, C \rightarrow E, D \rightarrow E \}$

Practice 4

$R(A, B, C, D, E, F)$

$F_1 = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, DE \rightarrow F, A \rightarrow E \}$

$F_2 = \{ A \rightarrow C, A \rightarrow E, C \rightarrow D, AE \rightarrow F, A \rightarrow B \}$

Given

$R(A, B, C, D, E)$

$F = \{ AB \rightarrow C, A \rightarrow B, C \rightarrow D, AD \rightarrow E \}$

Find the minimal cover.

Given

$R(A, B, C, D, E)$

$F = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C, C \rightarrow D, D \rightarrow E \}$

Find the minimal cover.

$R(A, B, C, D, E)$

$F = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C, C \rightarrow DE \}$

Find the minimal cover.

Practice 6

$R(A, B, C, D, E)$

$F = \{ A \rightarrow B, B \rightarrow C, AC \rightarrow D, D \rightarrow E, A \rightarrow E \}$

Find the minimal cover.

Given

$R(A, B, C, D, E)$

$F = \{ A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow D, D \rightarrow E \}$

Find the minimal cover.

Given

$R(A, B, C, D, E, F)$

$F = \{ A \rightarrow BC, B \rightarrow D, CD \rightarrow E, A \rightarrow D, E \rightarrow F \}$

Find the minimal cover.

Practice 4

$R(A, B, C, D, E)$

$F = \{ AB \rightarrow C, C \rightarrow D, D \rightarrow E, A \rightarrow B \}$

Find the minimal cover.

Practice 7

$R(A, B, C, D, E)$

$F = \{ AB \rightarrow C, A \rightarrow B, C \rightarrow D, A \rightarrow D, D \rightarrow E \}$

Find the minimal cover.